Build a count table

name _____

Copy one card into each column. The feature metadata is filled in for you.

samples (columns) →

sample metadata

card ID										
inflammation										

features (rows) ↓

feature metadata

feature	makes lactic acid?	encodes lysin?	counts									
O orange	yes	no										
G green	yes	yes										
B blue	no	yes										
P pink	no	no										
V vermillion	no	no										
S sky blue	no	no										
Y yellow	no	no										
X grey	no	no										

column total										
sum: makes lactic acid										
sum: encodes lysin										
mostly lactic acid? (Y/N)										

Questions

Work through these as a table. Everyone writes on their own sheet.

1. Column totals

Compare each column total with the n on the card front. Do they match?
Which card was measured least? Is its community different,
or just less well measured?

2. Collapse the rows

For one column, add "sum: makes lactic acid" and "sum: encodes lysin".
Why does that not equal the column total?

When is it safe to add rows up into groups, and when is it not?

3. Keep only high inflammation

Cross out every low-inflammation column in the counts.
What else on side 1 must you cross out so the tables still line up?

In R, sample metadata is a table with one ROW per sample, but the counts have one COLUMN per sample. Sketch your sample metadata the R way:

4. Report to the room

Count your cards into this table, then copy it onto the flip chart.

	inflammation	
	high	**low**
mostly lactic acid		
not mostly lactic acid		

n = _____ cards

Does your table suggest inflammation goes with the community?

Now look at the whole room on the flip chart. Does the room agree with your table? If not, who is wrong, and why?

5. Tomorrow

Your table has 8 rows and about 6 columns. Tomorrow's tables have 220 rows (bacteria) or 14,000 rows (genes), and hundreds of columns.
Which of today's steps would you never want to do by hand again?